SIS 3204Y(5) – Natural Language Processing

Week 7: Transformer Architecture, BERT and GPT

Examination Questions and Model Answers | Total: 30 marks | Suggested duration: 50–60 minutes

Part A – Examination Questions

Question 1 – Transformer Fundamentals (5 marks)

a) State two limitations of traditional recurrent neural networks (RNNs) that Transformers help address. (2 marks)

b) Explain why positional encoding is necessary in Transformer models. (2 marks) c)

State the purpose of residual connections in a Transformer layer. (1 mark)

Question 2 – Transformer Architecture (5 marks)

a) Explain the purpose of multi-head self-attention in a Transformer encoder. (2 marks) b)

Describe the role of the position-wise feed-forward network. (2 marks) c) State the purpose of

layer normalisation. (1 mark)

Question 3 – Understanding BERT (5 marks)

Consider the sentence: “The customer was satisfied with the service.” a)

Identify the Transformer architecture used by BERT. (1 mark)

b) Explain how BERT uses bidirectional context to represent the word “satisfied”. (2 marks)

c) Describe how BERT could be adapted to classify the sentiment of this sentence. (2 marks)

Question 4 – GPT and Text Generation (5 marks)

A GPT model receives the prompt: “The student opened the”. Next-token probabilities: book 0.60; door 0.25; laptop 0.10; window 0.05.

a) Identify the token selected using greedy decoding. (1 mark)

b) Explain what happens after this token is generated. (2 marks)

c) Explain why GPT uses causal (masked) self-attention. (2 marks)

Question 5 – Choosing Transformer Models (6 marks)

A software company is developing three NLP applications:

Application A: Identify names of people, organisations and locations in customer emails.

Application B: Generate a response to a customer's enquiry.

Application C: Translate customer emails from English into French.

For each application: a) Recommend an appropriate architecture (encoder-only, decoder only or encoder–decoder). (3 marks) b) Justify the choice based on language processing/generation. (3 marks)

Question 6 – Training, Inference and LLMs (4 marks)

a) Distinguish between training and inference in a Transformer model. (2 marks)

b) Explain the difference between a base GPT model trained for next-token prediction and an instruction-tuned LLM. (2 marks)

Part B – Model Answers and Marking Scheme

Question 1 – Transformer Fundamentals

Part	Expected answer	Marks
a	RNNs process tokens sequentially, limiting parallelisation.	1
a	RNNs may struggle to capture long-range dependencies.	1

b Self-attention alone does not explicitly encode token order.	1	
b Positional encoding supplies token position information.	1	
c Residual connections preserve information and improve gradient flow.	1	

Total: 5 marks

Question 2 – Transformer Architecture

Part	Expected answer	Marks
a Multiple attention heads operate in parallel.		1
a Heads can capture different syntactic/semantic relationships.		1
b The FFN independently transforms each token representation.		1
b Linear layers and non linear activation enrich representations.		1
c Layer normalisation normalises features and stabilises training.		1

Total: 5 marks

Question 3 – Understanding BERT

Part	Expected answer	Marks
a	Encoder-only Transformer.	1
b Bidirectional self-attention considers both left and right context.		1
b Produces a representation influenced by the whole sentence.		1
c Add a classification head, commonly using the [CLS]	1	

	representation.	
c Fine-tune on labelled sentiment examples to predict sentiment.		1

Total: 5 marks

Question 4 – GPT and Text Generation

Part	Expected answer	Marks
a	book (highest probability, 0.60).	1
b	Append the generated token to the sequence.	1
b	Predict the next token from the extended context; repeat until stopping.	1
c	Causal masking prevents attention to future positions.	1
c	It supports autoregressive prediction from preceding context only.	1

Total: 5 marks

Question 5 – Choosing Transformer Models

Part	Expected answer	Marks
A	Encoder-only (e.g. BERT).	1
A	Bidirectional contextual representations support entity tagging.	1
B	Decoder-only (e.g. GPT).	1
B	Autoregressive generation produces a response token by token.	1
C	Encoder–decoder (e.g. T5 or Marian).	1
C	Encoder represents source; decoder generates translation with causal self attention and cross attention.	1

Total: 6 marks

Question 6 – Training, Inference and LLMs

Part	Expected answer	Marks
a	Training updates parameters by optimising a loss.	1
a	Inference makes predictions with fixed learned parameters.	1
b	Base GPT is pretrained for next-token prediction.	1
b	Instruction tuning uses instruction–response examples to improve following requests.	1

Total: 4 marks

Marking Guidance

Award credit for equivalent technically correct explanations, not exact wording. For Question 5, accept alternative architectures or models when appropriately justified (e.g. decoder-only instruction-tuned LLMs for translation).

Assessment Coverage

Topic	Question	Marks
Transformer fundamentals and positional encoding	Q1	5
Transformer components	Q2	5
BERT and bidirectional understanding applications	Q3	5
GPT and autoregressive generation	Q4	5
Model selection and Training, inference and instruction-tuned LLMs	Q5 Q6	6 4
TOTAL		30

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